

RADIONUCLIDE SAFETY DATA SHEET					
RADIONUCLIDE: Ce-141			FORMS: Soluble		
PHYSICAL CHARACTERISTICS					
HALF-LIFE: 32.51 days					
DECAY EMISSIONS					
Gammas / X-rays		Betas / Positrons (+) / Electrons*		Alphas	
E (keV)	%	E (keV, Ave)	%	E (keV)	%
145	48	130	70		
36	14	181	30		
41	3	103*	19		
		139*	2		
- Only 4 most probable emissions per decay type included. Emissions below 10 keV or 1% excluded.					
STANFORD HAZARD CATEGORY					
C – level (low hazard): ≤ 20 mCi					
B – level (moderate hazard): > 20 mCi, ≤ 1 Ci					
A – level (high hazard): > 1 Ci					
EXTERNAL RADIATION HAZARDS			INTERNAL RADIATION HAZARDS		
Gamma dose rate, point source at 1 ft, 1 mCi: 0.47 mrem/h			Annual Limit on Intake: 2000 µCi (Ingestion) 600 µCi (Inhalation)		
Beta dose rate to skin, point source at 1 ft, 1 mCi: 212 mrem/h			The values above indicate the activity taken into the body that would result in either 5 rem to the whole body (CEDE) or 50 rem to an organ or tissue (CDE).		
Contamination skin dose, uniform deposit of 1 µCi per cm²: 6800 mrem/h					
SHIELDING			DOSIMETRY AND BIOASSAY REQS		
Gammas/X-rays: 1 mm of lead will reduce the gamma dose rate by 90%.			Whole-body and finger-ring dosimeters are required for handling 5 mCi or more, or 1 mCi amounts weekly . Urine assays may be required after large spills or contaminations.		
Betas/electrons: 1.5 mm of plastic will absorb all emissions.					
SPECIAL PROBLEMS AND PRECAUTIONS:					
1. Recommended survey probe: PGM					
2. Always wear protective gloves, a lab coat, and safety eyewear to protect the skin and eyes from contamination. Change gloves often.					
3. Survey work areas before, during, and after work. Work areas may require shielding to keep dose ALARA. Instrument and smear surveys are required.					
4. Segregate waste to those with half-lives of between 15 and 120 days . Survey the waste disposal area to ensure exposure rates are less than 2 mR/hr at 1 foot.					
5. Limit soluble waste to the sewer to less than 100 µCi/day per lab.					

References:

- Delacroix, D., Guerre, J.P., Leblanc, P., Hickman, C. (2002). Radionuclide and Radiation Protection Data Handbook (2nd ed.). Ashford, Kent: Nuclear Technology Publishing.
- Johnson, T.E., Birkby, B.K. (2012). Health Physics and Radiological Health (4th ed.). Baltimore, MD: Lippincott Williams & Wilkins.
- ICRP, 2008. Nuclear Decay Data for Dosimetric Calculations. ICRP Publication 107. Ann. ICRP 38 (3).
- Peplow, D. (2020) Specific Gamma-Ray Dose Constants with Current Emission Data. *Health Physics*, 118(4):402-416; 2020.
- Smith, D., Stabin, M. (2012) Exposure Rate Constants and Lead Shielding Values for Over 1,100 Radionuclides. *Health Physics*, 102(3): 271-291.
- 10.CFR.20 – Standards for Protection Against Radiation (2019). Retrieved from <https://www.nrc.gov/reading-rm/doc-collections/cfr/part020/>